“ Steering Cylinder Ends - Lubricate“ 134

Initial 500 Hours (for New Systems, Refilled Systems, and Converted Systems)

“ Cooling System Coolant Sample (Level 2) - Obtain“ 94

Every 500 Service Hours

“ Braking System - Test“ 85
“ Engine Oil and Filter - Change“. 107
“ Fuel System Primary Filter - Replace“ 116
“ Fuel Tank Strainer - Clean“ 118
“ Hydraulic System Oil Filter - Replace“ 124
“ Hydraulic System Oil Sample - Obtain“ 128
“ Isolation Mounts - Inspect“ 131

Every 1000 Service Hours

“ Battery - Clean/Check“ 80
“ Cooling System Pressure Cap - Clean/ Replace“ 95
“ Engine Mounts - Inspect“ 104
“ Engine Valve Lash - Check“ 109
“ Hydraulic Tank Breather - Replace“ 129
“ Hydraulic Tank Strainer - Clean“. 130
“ Rollover Protective Structure (ROPS) - Inspect“. 132

Every 2000 Service Hours

“ Engine Crankcase Breather - Replace“ 103

Every Year

“ Cooling System Coolant Sample (Level 2) - Obtain“ 94

Every 3000 Service Hours

“ Cooling System Water Temperature Regulator - Replace“ 96
“ Eccentric Weight Bearings - Inspect“ 100
“ Engine Water Pump - Inspect“ 109
“ Fuel Injector - Test/Change“ 110
“ Hydraulic System Oil - Change“ 121

Every 3 Years

“ Seat Belt - Replace“. 133

Every 6000 Service Hours or 3 Years

“ Cooling System Coolant Extender (ELC) - Add“ 89

Every 12 000 Service Hours or 6 Years

“ Cooling System Coolant (ELC) - Change“ 86

i08467606

Additive Spray System - Drain

SMCS Code: 5615-543

S/N: 6421–Up

S/N: 2C71–Up

S/N: BT71–Up

S/N: XGC1–Up

WARNING

Before service work is begun, park the machine on a hard level surface. Block the wheels of the machine so unexpected movement will not occur. Unexpected movement of the machine can cause personal injury or death.

NOTICE
Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting, and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, PERJ1017, “Dealer Service Tool Catalog” for tools and supplies suitable to collect and contain fluids on Cat ® products.

Dispose of all fluids according to local regulations and mandates.

NOTICE
Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

NOTICE
Not following the recommendations found in this manual can lead to reduced performance and compartment failure.

Prepare the machine for maintenance. Refer to “Prepare the Machine for Maintenance“ for more information.

1. Turn on the engine. Refer to “Engine Starting“ for more information.



Illustration 70 g06674306

(1) Additive spray system switch

2. Turn on the additive spray system switch (1). Refer to “Operator Controls“ for more information.

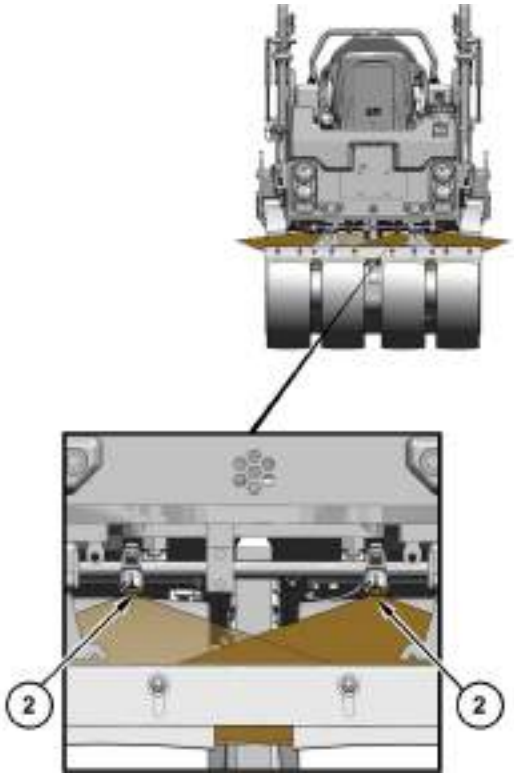


Illustration 71 g06674309

Rear view of the machine

(2) Nozzle

Note: Refer to “General Hazard Information“ for information pertains to containing fluid spillage.

3. Allow the tank to drain through the spray nozzles (2).

4. Place a suitable container under the spray nozzle (2). Loosen the locking collar on one of the spray nozzles (2) and allow the liquid to drain. Tighten the locking collar and ensure that the nozzle (2) is pointed towards the drum. Repeat for each nozzle (2).

Note: Dispose drained fluids according to local regulations.

5. Turn off the additive spray system. Refer to “Operator Controls“ for more information.

6. Shut down the engine. Refer to “Stopping the Engine“ for more information.

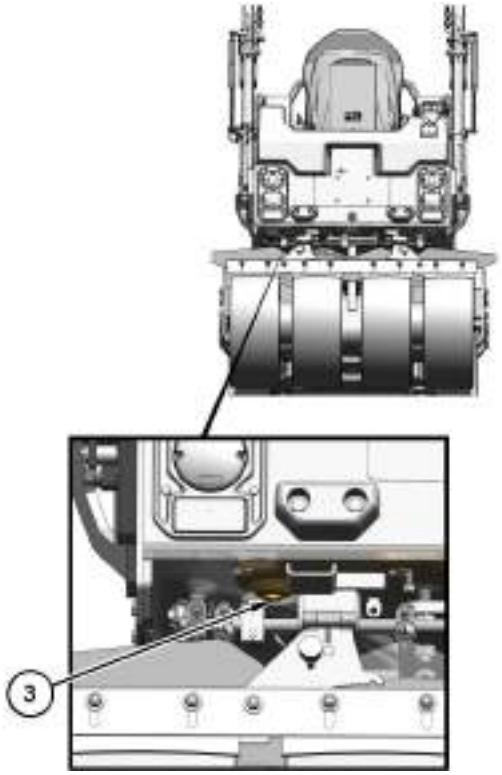


Illustration 72 g06674312

Rear view of the machine

(3) Drain plug

Note: Refer to “General Hazard Information“ for information pertains to containing fluid spillage.

Note: Dispose drained fluids according to local regulations.

7. To remove any remaining additive, place a suitable container under the additive tank drain plug (3) and remove the drain plug (3).

8. Allow the additive to drain, then reinstall the additive tank drain plug (3). Refer to Special Publications, SENR3130, “Torque Specifications” for more information.

Note: To refill the additive spray system tank. Refer to “Lubricant Viscosities“ and “Capacities (Refill)“ for more information.

Note: The additive spray system must be primed after the system is refilled. Refer to “Additive Spray System - Prime“ for information on priming the additive spray system.

i08467618

Additive Spray System - Prime

SMCS Code: 5615-548

S/N: 6421–Up

S/N: 2C71–Up

S/N: BT71–Up

S/N: XGC1–Up

⚠ WARNING

To avoid personal injury, always wear eye and face protection when using pressurized air.

⚠ WARNING

Before service work is begun, park the machine on a hard level surface. Block the wheels of the machine so unexpected movement will not occur. Unexpected movement of the machine can cause personal injury or death.

NOTICE

Not following the recommendations found in this manual can lead to reduced performance and compartment failure.

NOTICE
Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting, and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, PERJ1017, “Dealer Service Tool Catalog” for tools and supplies suitable to collect and contain fluids on Cat ® products.

Dispose of all fluids according to local regulations and mandates.

NOTICE
Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

Prepare the machine for maintenance. Refer to “Prepare the Machine for Maintenance“ for more information.

Perform the following steps to prime the additive spray system:

1. Turn on the engine. Refer to “Engine Starting“ for more information.



Illustration 73 g06612836

(1) Cap

2. Remove the cap (1) from the additive spray system tank.



Illustration 74 g06612840

(2) Additive spray system switch

3. Turn the additive spray system switch (2) to ON position. Refer to “Operator Controls“ for more information. Run the additive spray system until the system begins to spray the liquid.

4. Turn the additive spray system switch (2) to OFF position. Refer to “Operator Controls“ for more information.

5. Reinstall the cap (1) on the additive spray system tank.

6. Shut down the engine. Refer to “Stopping the Engine“ for more information.

i08612480

Backup Alarm - Test

SMCS Code: 7406-081

WARNING

Before service work is begun, park the machine on a hard level surface. Block the wheels of the machine so unexpected movement will not occur. Unexpected movement of the machine can cause personal injury or death.

NOTICE
Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

Prepare the machine for maintenance. Refer to “Prepare the Machine for Maintenance“ for more information.

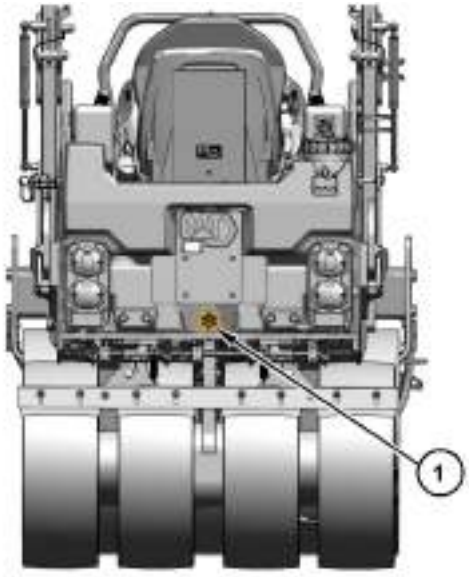


Illustration 75 g06674415

Rear view of the machine

(1) Backup alarm

Backup alarm (1) is at the rear of the machine.

1. Engage the parking brake. Refer to “Operator Controls“ for more information.

2. Turn the engine start switch key to the ON position. Do not start the engine.

3. Move the propel control lever to the REVERSE position. The backup alarm (1) should sound immediately. The backup alarm (1) will continue to sound until the propel control lever is moved to the NEUTRAL position or to the FORWARD position. Refer to “Operator Controls“ for more information.

Note: Consult your cat ® dealer. If the backup alarm (1) does not sound, make the necessary repairs before operating the machine.

i08467706

Battery - Clean/Check

SMCS Code: 1401-535; 1401-070; 1402-535; 1402-070

⚠ WARNING

Before service work is begun, park the machine on a hard level surface. Block the wheels of the machine so unexpected movement will not occur. Unexpected movement of the machine can cause personal injury or death.

NOTICE
Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

Prepare the machine for maintenance. Refer to “Prepare the Machine for Maintenance“ for more information.

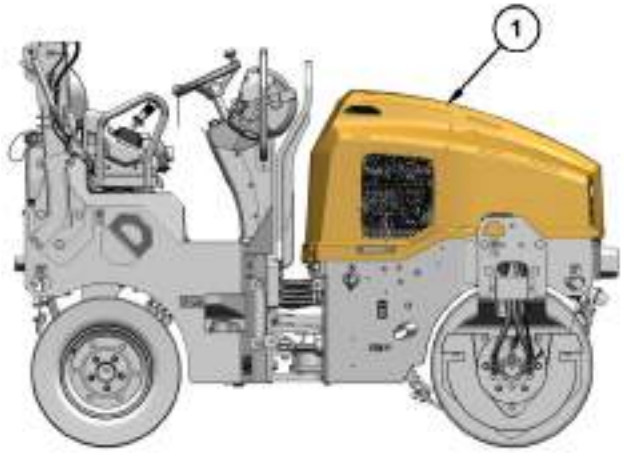


Illustration 76 g06674584

(1) Hood

To access the battery, open hood (1) assembly. Refer to “Access Doors and Covers“ for more information.

Note: The batteries supplied with the machine are maintenance free batteries. You do not need to check the level of electrolyte in the maintenance free batteries.

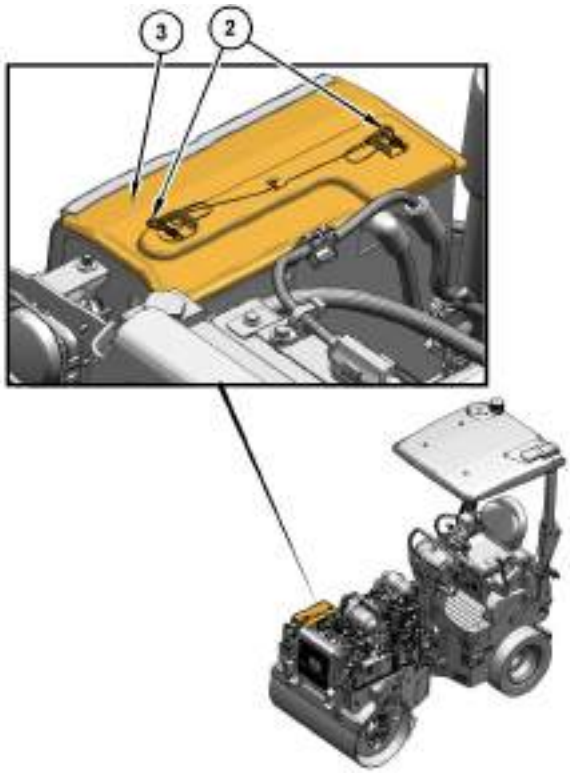


Illustration 77 g06674622

(2) Lock pin
(3) Top cover

Remove lock pins (2) and top cover (3) to get access to the battery.

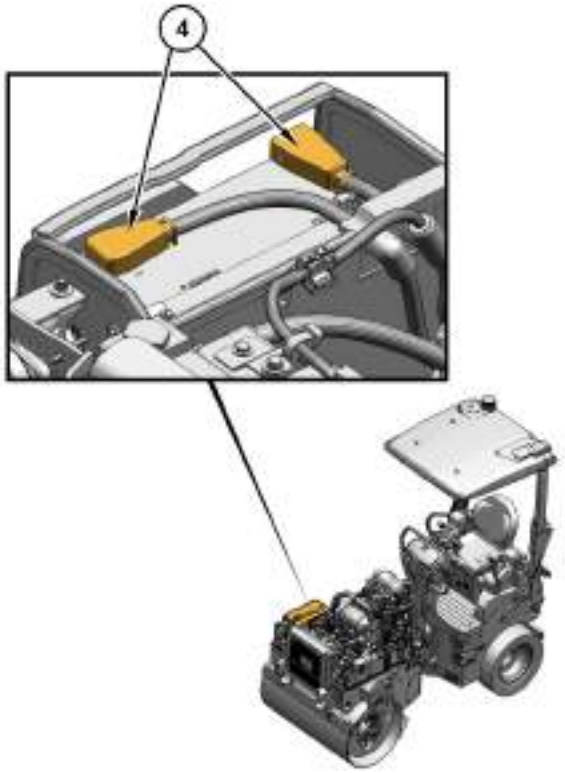


Illustration 78 g06674626

(4) Terminal

- Clean the top of the batteries with clean cloth.
- Clean the battery terminals (4). If required, coat the battery terminals with petroleum jelly.

i08316356

Battery - Recycle

SMCS Code: 1401-561

Always recycle a battery. Never discard a battery.

Always return used batteries to one of the following locations:

- A battery supplier
- An authorized battery collection facility
- Recycling facility

i08467901

Battery or Battery Cable - Inspect/Replace

SMCS Code: 1401-040; 1401-510; 1402-040; 1402-510

WARNING

To avoid personal injury from electrical shock, ensure that the negative battery cable is disconnected first.

WARNING

Before service work is begun, park the machine on a hard level surface. Block the wheels of the machine so unexpected movement will not occur. Unexpected movement of the machine can cause personal injury or death.

NOTICE
Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

NOTICE
Not following the recommendations found in this manual can lead to reduced performance and compartment failure.

Prepare the machine for maintenance. Refer to “Prepare the Machine for Maintenance“ for more information.

1. Turn all switches to the OFF position. Refer to “Operator Controls“ for more information.
2. Turn the engine start switch to the OFF position. Refer to “Stopping the Engine“ for more information.
3. Turn the battery disconnect switch to OFF position. Refer to “Battery Disconnect Switch“ for more information.

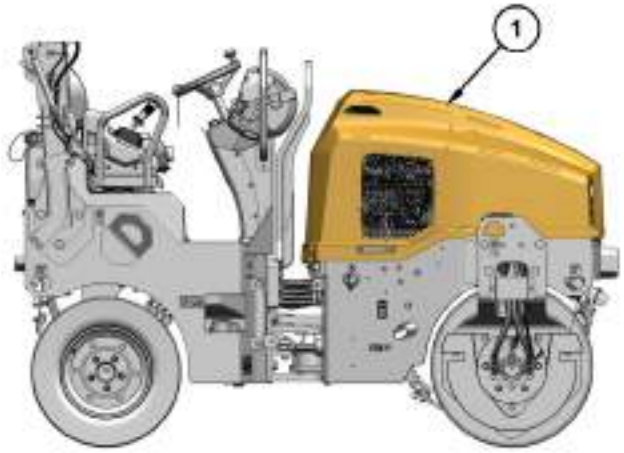


Illustration 79 g06674584

(1) Hood

4. Open the engine compartment hood (1) to access the battery or battery cables. Refer to “Access Doors and Covers“ for more information.

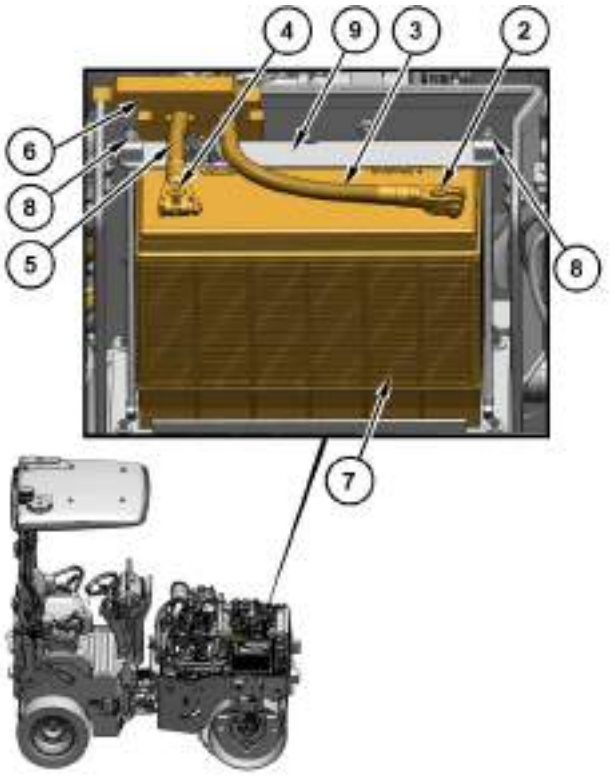


Illustration 80 g06674903

(2) Negative battery terminal
(3) Negative battery cable
(4) Positive battery terminal
(5) Positive battery cable
(6) Foam
(7) Battery
(8) Bolt
(9) Retrainer plate

5. Disconnect negative battery cable (3) from negative battery terminal (2) .

6. Disconnect positive battery cable (5) from positive battery terminal (4).

7. Remove foam (6) to remove battery cables.

8. Remove positive battery cable (5) from the engine starter motor.

9. Loosen bolts (8) and retrainer plate (9) to remove battery (7).

10. Perform the necessary repairs. If required replace the battery (7) or the cables.

11. Connect positive battery cable (5) to engine starter motor.

12. Reconnect positive battery cable (5) to positive battery terminal (4) .

13. Reconnect negative battery cable (3) to negative battery terminal (2).

14. Reinstall foam (6) and retainer plate (9) and bolts (8) to secure battery (7) and battery cables.

15. Turn the battery disconnect switch to (On) position. Refer to “Battery Disconnect Switch“ for more information.

16. Close the engine compartment hood (1). Refer to “Access Doors and Covers“ for more information.

i08467946

Belts - Inspect/Adjust/Replace

SMCS Code: 1357-040; 1357-025; 1357-510

WARNING

Personal injury can result from rotating and moving parts. Stay clear of all rotating and moving parts.

Never attempt adjustments while the machine is moving or the engine is running unless otherwise specified.

The machine must be parked on a level surface and the engine stopped.

Attach a “Do Not Operate” or similar warning tag to the start switch and controls before servicing, repairing, or making adjustments to the machine.

WARNING

Accidental machine starting can cause injury or death to personnel working on the machine.

To avoid accidental machine starting, turn the battery disconnect switch to the OFF position and remove the key. If the machine is not equipped with a battery disconnect switch, disconnect the battery cables from the battery and tape the battery clamps.

Place a do not operate tag at the battery disconnect switch location to inform personnel that the machine is being worked on.

NOTICE
Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

NOTICE
Not following the recommendations found in this manual can lead to reduced performance and compartment failure.

Prepare the machine for maintenance. Refer to “Prepare the Machine for Maintenance“ for more information.

Your engine is equipped with one belt that operates the alternator and the water pump. For maximum engine performance and maximum utilization of your engine, inspect the belt for wear and for cracking. Check the tension of the belts. Adjust the tension of the belt to minimize belt slippage. Belt slippage will decrease the belt life. Belt slippage will also cause poor performance.

Alternator Belt and Water Pump Belt

1. Turn the engine to OFF position. Refer to “Stopping the Engine“ for more information.

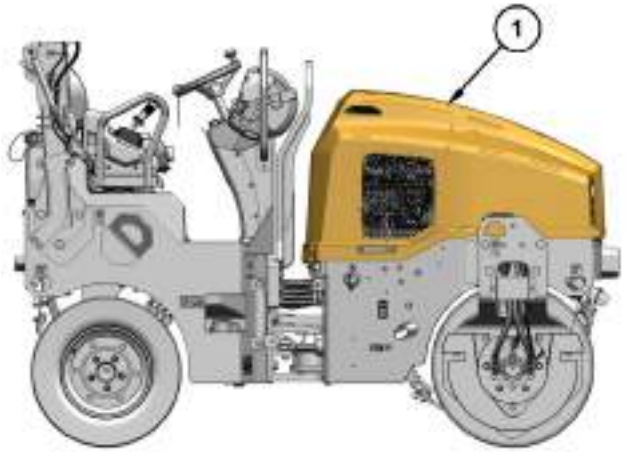


Illustration 81 g06674584

(1) Hood

2. Open hood (1) to get access to the engine. Refer to “Access Doors and Covers“ for more information.

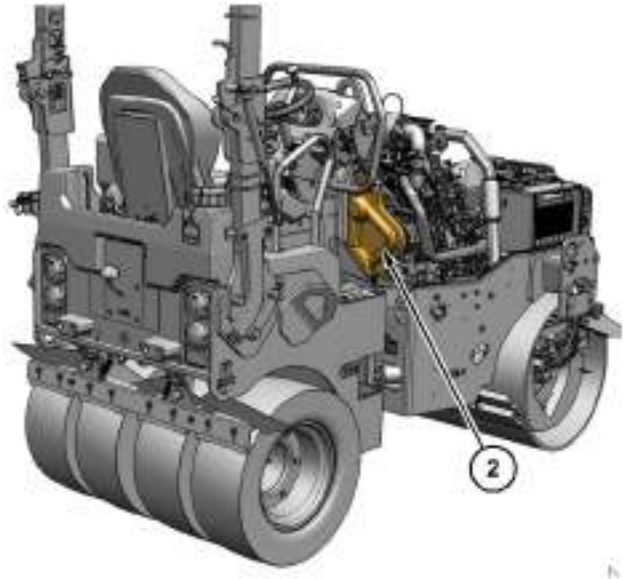


Illustration 82 g06675029

(2) Belt guard

3. Remove belt guard (2) to gain access to the belt.



Illustration 83 g06674988

(3) Bolt
(4) Alternator
(5) Belt

4. To adjust belt (5), loosen mounting bolts (3).

5. To achieve the correct adjustment, move alternator (4) inward or outward, as required.

6. Tighten mounting bolts (3).

Note: The alternator shaft nut must be tightened to a torque of 50 ± 5 N·m (36.8 ± 3.7 lb ft).

7. If a new belt (5) is installed, check belt (5) adjustment again after thirty minutes of engine operation at the rated speed. Adjust the belt tension again if the tension has changed.

Note: The correct tension for new belt (5) is 400 N·m (295 lb ft) to 490 N·m (361 lb ft). The correct belt tension for used belt (5) is 267 N·m (197 lb ft) to 356 N·m (263 lb ft) that has been in operation for thirty minutes or more at the rated speed.

8. Install the belt guard (2).

9. Close hood (1) of the engine compartment. Refer to “Access Doors and Covers“ for more information.

i04991551

Braking System - Test

SMCS Code: 4250-081; 4267-081

Note: If the machine configuration changes, the parking brakes must be tested.

Check the area around the machine. Make sure that the machine is clear of personnel and clear of obstacles.

Put the steering frame lock in the UNLOCKED position.

Fasten the seat belt before checking the parking brake.

The following tests are used to determine if the parking brake is functional on a specified grade or a specified slope. These tests are not intended to measure the maximum brake holding effort. Read all of the steps before you perform the following procedure.

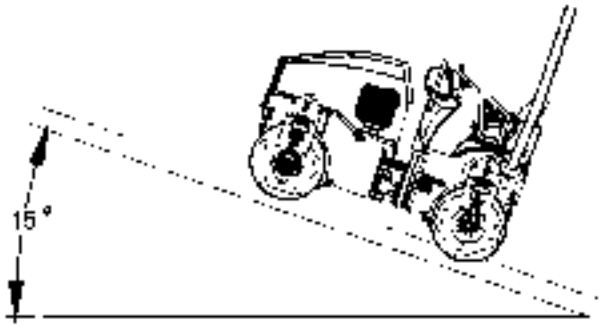


Illustration 84 g03170759

Position the machine on the incline of the slope, but near the base of the slope in order to check the parking brake. The test position should be 26 percent or a 15 degree slope.

1. Start the engine. Refer to the Operation and Maintenance Manual, “Engine Starting” for information on starting the engine.

2. Move the machine into the test position.

3. Place the throttle control into the LOW IDLE position.

4. Engage the parking brake.

The machine should not move under the following conditions.

- The engine is at low idle.
- The parking brake is applied.
- The machine is positioned on the specified slope.

⚠ WARNING

Personal injury can result if the machine moves while testing.

If the machine begins to move, release the parking brake and use the propel lever in order to move the machine to a level surface.

5. Park the machine on a level surface.

6. Stop the engine.

NOTICE
If the machine moved during the brake test, consult your Caterpillar dealer.

The dealer must inspect the brake system and make any necessary repairs before the machine is returned to operation.

i08468079

Cooling System Coolant (ELC) - Change

SMCS Code: 1395-044-NL

⚠ WARNING

Personal injury can result from hot coolant, steam and alkali.

At operating temperature, engine coolant is hot and under pressure. The radiator and all lines to heaters or the engine contain hot coolant or steam. Any contact can cause severe burns.

Remove cooling system pressure cap slowly to relieve pressure only when engine is stopped and cooling system pressure cap is cool enough to touch with your bare hand.

Do not attempt to tighten hose connections when the coolant is hot, the hose can come off causing burns.

Cooling System Coolant Additive contains alkali. Avoid contact with skin and eyes.

⚠ WARNING

Personal injury can result from rotating and moving parts. Stay clear of all rotating and moving parts.

Never attempt adjustments while the machine is moving or the engine is running unless otherwise specified.

The machine must be parked on a level surface and the engine stopped.

Attach a "Do Not Operate" or similar warning tag to the start switch and controls before servicing, repairing, or making adjustments to the machine.

NOTICE
Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting, and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, PERJ1017, "Dealer Service Tool Catalog" for tools and supplies suitable to collect and contain fluids on Cat ® products.

Dispose of all fluids according to local regulations and mandates.

NOTICE
Mixing Extended Life Coolant (ELC) with other products reduces the effectiveness of the coolant and shortens coolant life. Use only Caterpillar products or commercial products that have passed the Caterpillar EC-1 specifications for premixed or concentrate coolants. Use only Caterpillar Extender with Caterpillar ELC. Failure to follow these recommendations could result in the damage to cooling systems components.

If ELC cooling system contamination occurs, refer to Operation and Maintenance, "Extended Life Coolant (ELC)" under the topic ELC Cooling System Contamination.

NOTICE
Do not change the coolant until you read and understand the material in the Cooling System Specifications section.

NOTICE
Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

NOTICE
Not following the recommendations found in this manual can lead to reduced performance and compartment failure.

Prepare the machine for maintenance. Refer to "Prepare the Machine for Maintenance" for more information.

Drain the coolant whenever the coolant is dirty. Drain the coolant when foam is observed.

1. Stop the engine. Refer to "Stopping the Engine" for more information.

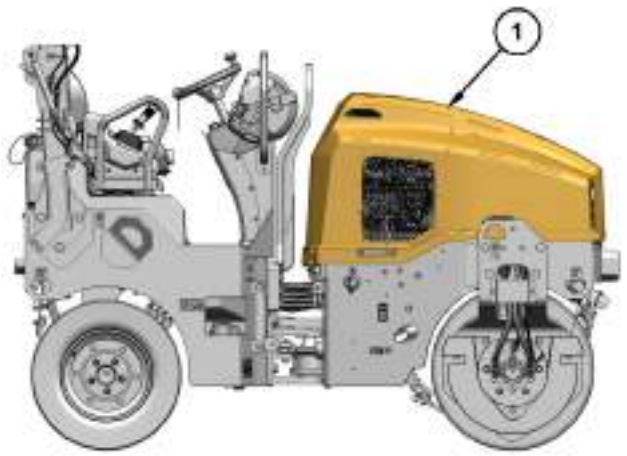


Illustration 85 g06674584

(1) Hood

2. Open the engine compartment hood (1). Refer to the “Access Doors and Covers“ for more information. Allow the cooling system to cool completely.

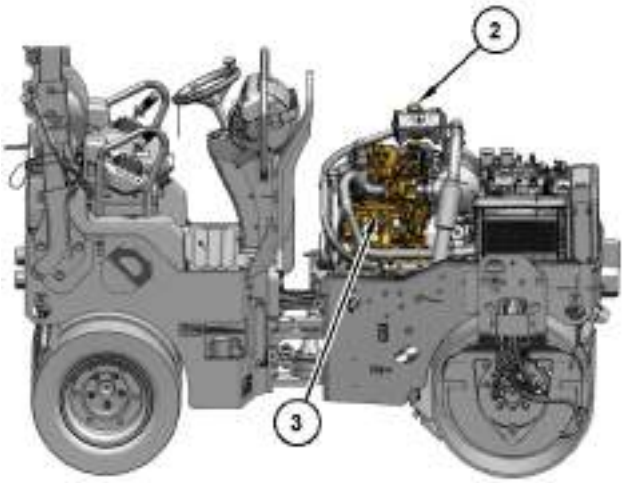


Illustration 86 g06675077

(2) Pressure cap
(3) Engine

3. Slowly loosen the cooling system pressure cap (2) to relieve system pressure. Refer to “System Pressure Release“ for more information. Remove the cooling system pressure cap (2).

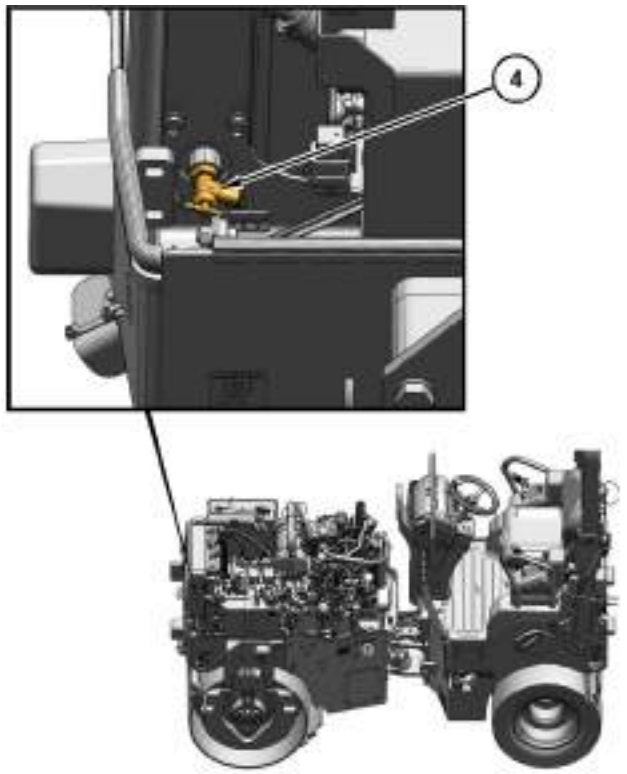


Illustration 87 g06675079

parts removed for better clarity

(4) Shut off valve

4. The radiator drain shut off valve (4) can be accessed through the engine compartment. The drain hose for the hydraulic system can also be swapped for cooling system.

Note: Refer to “General Hazard Information“ for more information that pertains to containing fluid spillage.

5. Open shut off valve (4) to drain the coolant completely. Collect the coolant in a suitable container.

Note: Dispose the coolant according to local regulations.

6. Close shut off valve (4).

7. Fill the cooling system with a solution of clean water and/or cooling system cleaner. The concentration of the cleaner in the system should be between 6 percent to 10 percent.

8. Install the cooling system pressure cap (2).

9. Close the engine compartment hood (1). Refer to the “Access Doors and Covers“ for more information.

10. Turn on engine (3). Refer to “Engine Starting“ for more information. Run engine (3) for 90 minutes.

11. Turn off engine (3). Refer to “Stopping the Engine“ for more information. Allow the cooling system to cool completely .

12. Open the engine compartment hood (1). Refer to the “Access Doors and Covers“ for more information.

13. Remove the cooling system pressure cap (2). Refer to “System Pressure Release“ for more information.

Note: Refer to “General Hazard Information“ for more information that pertains to containing fluid spillage.

14. The radiator drain shut off valve (4) can be accessed through the engine compartment. Drain the fluid completely.

Note: Dispose the fluid according to local regulations.

Note: Refer to “General Hazard Information“ for more information that pertains to containing fluid spillage.

15. Flush the cooling system with water until the draining water is transparent.

Note: Dispose the fluid according to local regulations.

16. Close radiator shut off valve (4).

17. Refill the cooling system. Refer to “Capacities (Refill)“ for the proper amount.

18. Turn on engine (3) and run the engine. Refer to “Engine Starting“ for more information. Remove the filler cap until the thermostat opens and the coolant level stabilizes.

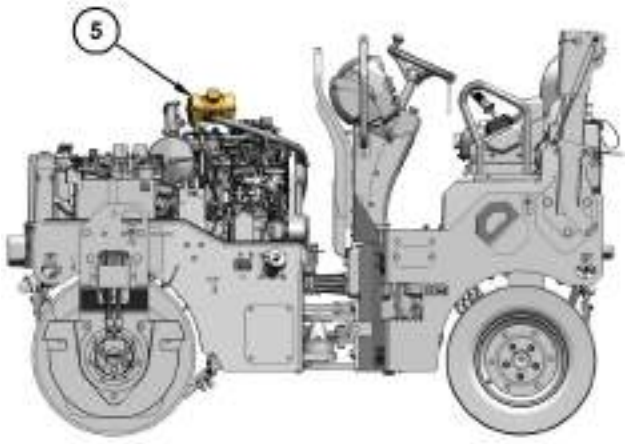


Illustration 88 g06675105

Parts removed for better clarity

(5) Shunt tank

19. Maintain the coolant level between the indicated lines on shunt tank (5).

20. Inspect the gasket on the cooling system pressure cap (2). Replace the cooling system pressure cap (2) if the gasket is damaged.

21. Install the cooling system pressure cap (2).

22. Close the engine compartment hood (1). Refer to the “Access Doors and Covers“ for more information.

i08468117

Cooling System Coolant Extender (ELC) - Add

SMCS Code: 1352-544-NL

WARNING

At operating temperature, the engine coolant is hot and under pressure.

Steam can cause personal injury.

Check the coolant level only after the engine has been stopped and the fill cap is cool enough to touch with your bare hand.

Remove the fill cap slowly to relieve pressure.

Cooling system conditioner contains alkali. Avoid contact with the skin and eyes to prevent personal injury.

WARNING

Pressurized System: Hot coolant can cause serious burns. To open the cooling system filler cap, stop the engine and wait until the cooling system components are cool. Loosen the cooling system pressure cap slowly in order to relieve the pressure.

NOTICE
Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting, and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, PERJ1017, “Dealer Service Tool Catalog” for tools and supplies suitable to collect and contain fluids on Cat ® products.

Dispose of all fluids according to local regulations and mandates.

NOTICE
Topping off or mixing Cat ELC with other products that do not meet Caterpillar EC-1 specifications reduces the effectiveness of the coolant, shortens coolant service life, and may cause premature wear to components.

Use only Caterpillar products or commercial products that have passed the Caterpillar EC-1 specification for pre-mixed or concentrate coolants. Use only Extender with Cat ELC.

Failure to follow these recommendations can result in shortened cooling system component life.

NOTICE
Mixing Extended Life Coolant (ELC) with other products reduces the effectiveness of the coolant and shortens coolant life. Use only Caterpillar products or commercial products that have passed the Caterpillar EC-1 specifications for premixed or concentrate coolants. Use only Caterpillar Extender with Caterpillar ELC. Failure to follow these recommendations could result in the damage to cooling systems components.

If ELC cooling system contamination occurs, refer to Operation and Maintenance, “Extended Life Coolant (ELC)” under the topic ELC Cooling System Contamination.

NOTICE
Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

NOTICE
Do not change the coolant until you read and understand the material in the Cooling System Specifications section.

NOTICE
Not following the recommendations found in this manual can lead to reduced performance and compartment failure.

Prepare the machine for maintenance. Refer to “Prepare the Machine for Maintenance“ for more information.

1. Stop the engine. Refer to “Stopping the Engine“ for more information. Allow the cooling system to cool completely .

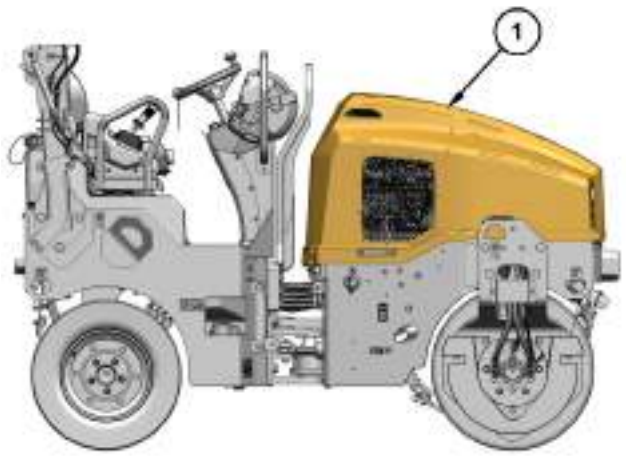


Illustration 89 g06674584

(1) Hood

2. Open the engine compartment hood (1). Refer to “Access Doors and Covers“ for more information.

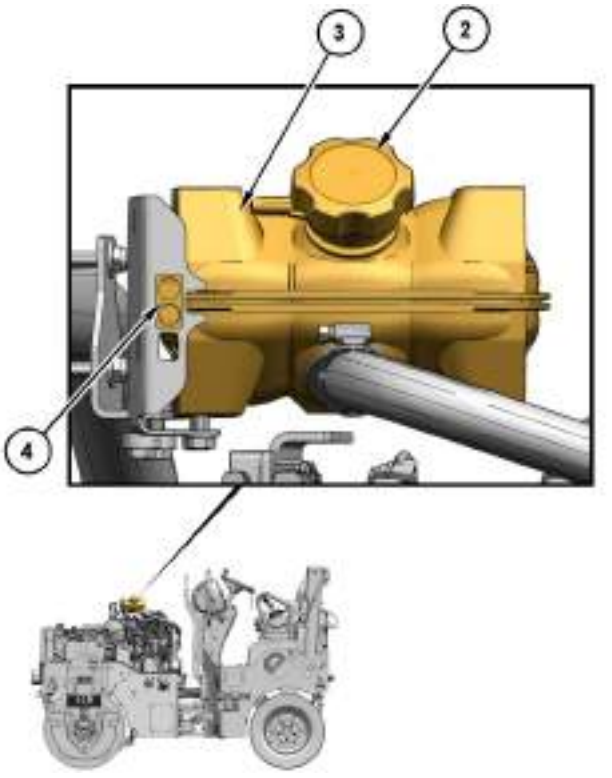


Illustration 90 g06675124

(2) Pressure cap
(3) Shunt tank
(4) Film

3. Slowly loosen the cooling system pressure cap (2) to relieve system pressure. Remove the cooling system pressure cap (2). Refer to “System Pressure Release“ for more information.

Note: Refer to “General Hazard Information“ for more information that pertains to containing fluid spillage.

4. If required, drain enough coolant from the radiator to allow the addition of the Extender.

Note: Dispose fluids according to local regulations.

5. Add the recommended amount of extender to the coolant system. Refer to the Special Publication, SEBU6250, “Extended Life Coolant (ELC)” for the proper amount.

Note: Refer to “Capacities (Refill)“ for more information.

6. Maintain the coolant level between the indicated lines on shunt tank (3).

7. Inspect the gasket on the cooling system pressure cap (2). Replace the cooling system pressure cap (2) if the gasket is damaged.

8. Install the cooling system pressure cap (2).

9. Close the engine compartment hood (1). Refer to “Access Doors and Covers“ for more information.

For additional information on the addition of extender, see Special Publication, SEBU6250, “Coolant Recommendations” or consult your Cat ® dealer.

i08468251

Cooling System Coolant Level - Check

SMCS Code: 1350-535-FLV

⚠ WARNING

At operating temperature, the engine coolant is hot and under pressure.

Steam can cause personal injury.

Check the coolant level only after the engine has been stopped and the fill cap is cool enough to touch with your bare hand.

Remove the fill cap slowly to relieve pressure.

Cooling system conditioner contains alkali. Avoid contact with the skin and eyes to prevent personal injury.

⚠ WARNING

Pressurized System: Hot coolant can cause serious burns. To open the cooling system filler cap, stop the engine and wait until the cooling system components are cool. Loosen the cooling system pressure cap slowly in order to relieve the pressure.

NOTICE
Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting, and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, PERJ1017, “Dealer Service Tool Catalog” for tools and supplies suitable to collect and contain fluids on Cat ® products.

Dispose of all fluids according to local regulations and mandates.

NOTICE
Do not change the coolant until you read and understand the material in the Cooling System Specifications section.

NOTICE
Do not change the coolant until you read and understand the material in the Cooling System Specifications section.

NOTICE
Mixing Extended Life Coolant (ELC) with other products reduces the effectiveness of the coolant and shortens coolant life. Use only Caterpillar products or commercial products that have passed the Caterpillar EC-1 specifications for premixed or concentrate coolants. Use only Caterpillar Extender with Caterpillar ELC. Failure to follow these recommendations could result in the damage to cooling systems components.

If ELC cooling system contamination occurs, refer to Operation and Maintenance, “Extended Life Coolant (ELC)” under the topic ELC Cooling System Contamination.

Note: To access the areas required to perform, a portable access system (ladder, stair assembly, man lift, or other portable access system) that is suitable and compliant to local regulations may be necessary.

Prepare the machine for maintenance. Refer to “Prepare the Machine for Maintenance“ for more information.

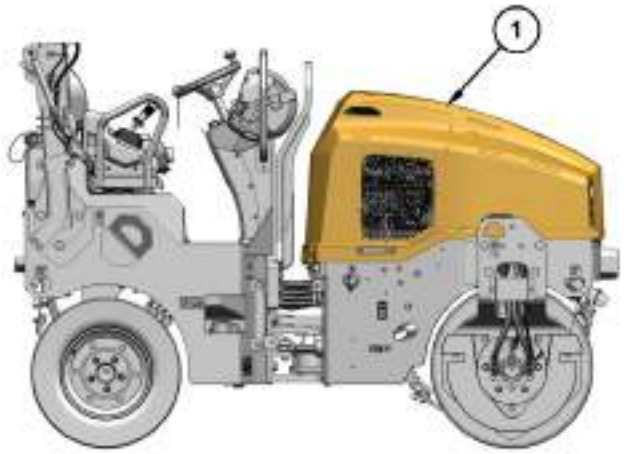


Illustration 91 g06674584

(1) Hood

1. Open the engine compartment hood (1). Refer to “Access Doors and Covers“ for more information.

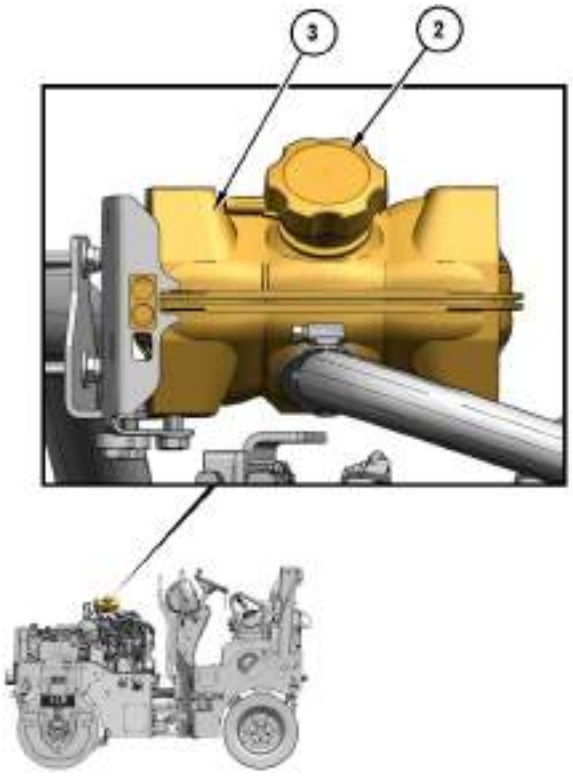


Illustration 92 g06675368

(2) Pressure cap
(3) Shunt tank

2. Slowly loosen the pressure cap (2) on the shunt tank (3) to release pressure in the system. Slowly remove the pressure cap (2) on shunt tank (3). Refer to “System Pressure Release“ for more information.

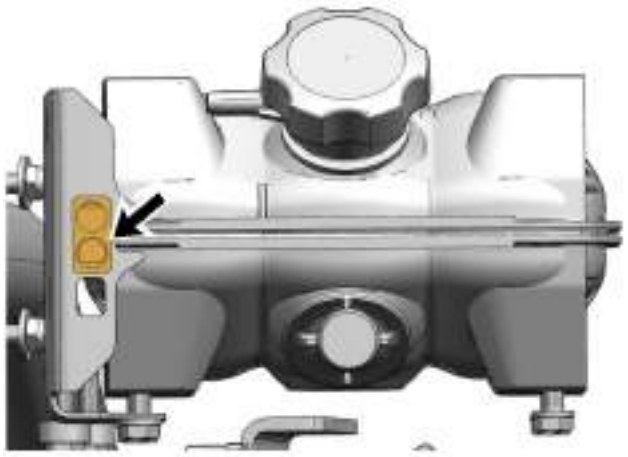


Illustration 93 g06675306
Exploded view of shunt tank

3. Maintain the coolant level between the indicated lines. If coolant must be added daily, check the cooling system for leaks. Repair the leaks as the leaks are found.

4. Inspect the gasket. Inspect pressure cap (2) on shunt tank (3). Clean the pressure cap (2) on the shunt tank (3) with a clean cloth. Replace the cap if the gasket or the cap are damaged.

5. Install pressure cap (2) on shunt tank (3).

6. Close the engine compartment hood (1). Refer to “Access Doors and Covers“ for more information.

i08468296

Cooling System Coolant Sample (Level 1) - Obtain

SMCS Code: 1350-008; 1395-554; 1395-008; 7542; 7542-008

⚠ WARNING

Personal injury can result from hot coolant, steam and alkali.

At operating temperature, engine coolant is hot and under pressure. The radiator and all lines to heaters or the engine contain hot coolant or steam. Any contact can cause severe burns.

Remove cooling system pressure cap slowly to relieve pressure only when engine is stopped and cooling system pressure cap is cool enough to touch with your bare hand.

Do not attempt to tighten hose connections when the coolant is hot, the hose can come off causing burns.

Cooling System Coolant Additive contains alkali. Avoid contact with skin and eyes.

⚠ WARNING

Pressurized System: Hot coolant can cause serious burns. To open the cooling system filler cap, stop the engine and wait until the cooling system components are cool. Loosen the cooling system pressure cap slowly in order to relieve the pressure.

NOTICE
Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting, and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, PERJ1017, “Dealer Service Tool Catalog” for tools and supplies suitable to collect and contain fluids on Cat ® products.

Dispose of all fluids according to local regulations and mandates.

NOTICE
Always use a designated pump for oil sampling, and use a separate designated pump for coolant sampling. Using the same pump for both types of samples may contaminate the samples that are being drawn. This contaminate may cause a false analysis and an incorrect interpretation that could lead to concerns by both dealers and customers.

NOTICE
Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

NOTICE
Not following the recommendations found in this manual can lead to reduced performance and compartment failure.

Prepare the machine for maintenance. Refer to “Prepare the Machine for Maintenance“ for more information.

Note: It is not necessary to obtain a coolant sample (level 1). If the cooling system is filled with Cat ELC (Extended Life Coolant). cooling systems that are filled with Cat ELC should have a coolant sample (level 2) that is obtained at the recommended interval.

Note: Obtain a coolant sample (level 1) if the cooling system is filled with any other coolant instead of Cat ELC. The following types of coolants are included.

- Commercial long life coolants that meet the Caterpillar Engine Coolant Specification -1 (Caterpillar EC-1)
- Cat ® Diesel Engine Antifreeze/Coolant (DEAC)
- Commercial heavy-duty antifreeze/coolant solution

Note: Level 1 results may indicate a need for level 2 analysis.

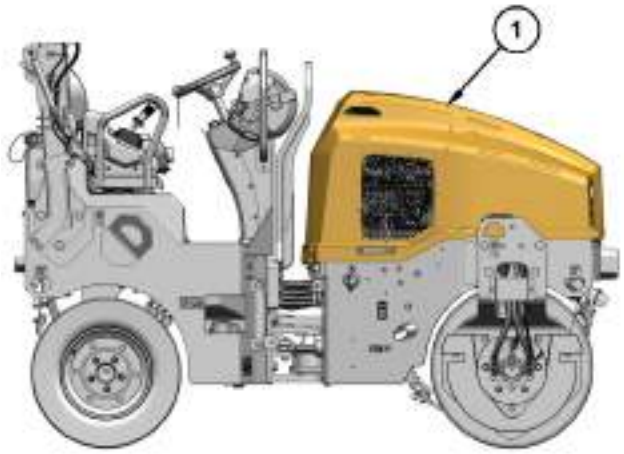


Illustration 94 g06674584

(1) Hood

1. Open the engine compartment hood (1). Refer to “Access Doors and Covers“ for more information.
2. Start the engine momentarily to circulate the coolant again. Refer to “Engine Starting“ for more information.
3. Shut off the engine. Refer to “Stopping the Engine“ for more information.

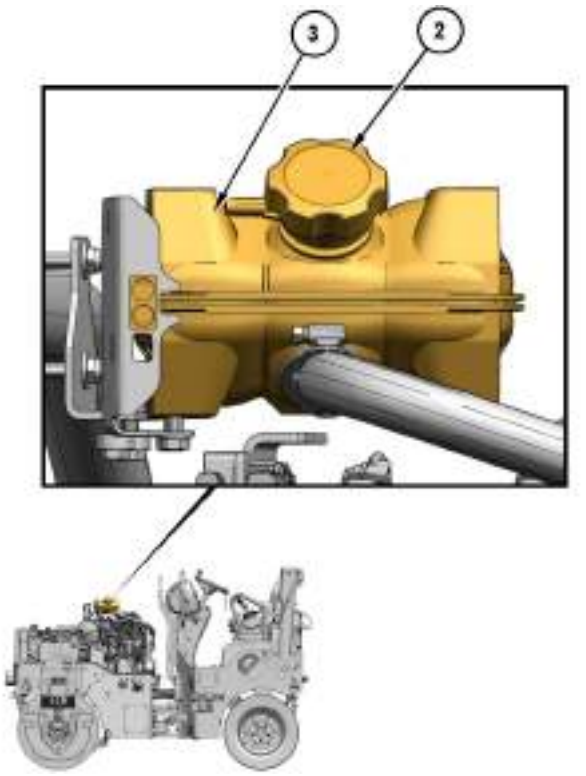


Illustration 95 g06675368

(2) Pressure cap
(3) Shunt tank

4. Slowly loosen the cooling system pressure cap (2) to release the system pressure. Refer to “System Pressure Release“ for more information. Remove pressure cap (2) from shunt tank (3).

5. Use a vacuum pump to obtain the sample from shunt tank (3). Refer to Special Publications, PEHP6001, “How To Take A Good Oil Sample” for more information.

6. Obtain the sample of the coolant as close as possible recommend in the sampling interval.

To receive the full effect of scheduled oil sampling analysis, a consistent trend of data must be established. To establish a pertinent history of data, perform consistent samplings that are evenly spaced. Supplies for collecting samples can be obtained from the Cat ® dealer.

7. Install the cooling system pressure cap (2) on shunt tank (3).

8. Close the engine compartment hood (1). Refer to “Access Doors and Covers“ for more information.

Use the following guidelines for proper sampling of the coolant:

- Complete the information on the label for the sampling bottle before you begin to take the samples.
- Keep the unused sampling bottles stored in plastic bags.
- Keep the lids on empty sampling bottles until you are ready to collect the sample.
- Place the sample in the mailing tube immediately after obtaining the sample to avoid contamination.
- Never collect samples from the drain for a system.

Note: For additional information about coolant analysis. Refer to Special Publications, SEBU6250, “Machine Fluid Recommendations” or consult your Cat ® dealer.

i07349178

Cooling System Coolant Sample (Level 2) - Obtain

SMCS Code: 1350-008; 1395-554; 1395-008; 7542; 7542-008

Reference: Refer to Operation and Maintenance Manual, “Cooling System Coolant Sample (Level 1) - Obtain” for the guidelines for proper sampling of the coolant.

Obtain the sample of the coolant as close as possible to the recommended sampling interval. Supplies for collecting samples can be obtained from your Cat dealer.

Submit the sample for Level 2 analysis.

Reference: For additional information about coolant analysis, refer to Special Publication, SEBU6250, “Caterpillar Machine Fluids Recommendations” or consult your Cat dealer.

i08468367

Cooling System Pressure Cap - Clean/Replace

SMCS Code: 1382-070; 1382-510

WARNING

Personal injury can result from hot coolant, steam and alkali.

At operating temperature, engine coolant is hot and under pressure. The radiator and all lines to heaters or the engine contain hot coolant or steam. Any contact can cause severe burns.

Remove cooling system pressure cap slowly to relieve pressure only when engine is stopped and cooling system pressure cap is cool enough to touch with your bare hand.

Do not attempt to tighten hose connections when the coolant is hot, the hose can come off causing burns.

Cooling System Coolant Additive contains alkali. Avoid contact with skin and eyes.

WARNING

Pressurized System: Hot coolant can cause serious burns. To open the cooling system filler cap, stop the engine and wait until the cooling system components are cool. Loosen the cooling system pressure cap slowly in order to relieve the pressure.

NOTICE
Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

NOTICE
Not following the recommendations found in this manual can lead to reduced performance and compartment failure.

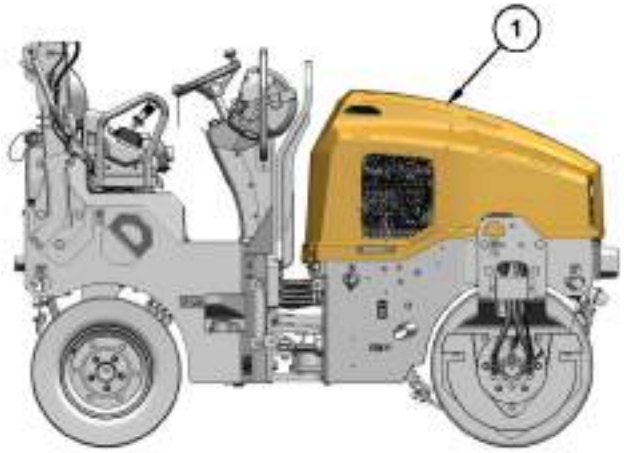


Illustration 96 g06674584

(1) Hood

1. Open the engine compartment hood (1). Refer to “Access Doors and Covers“ for more information.

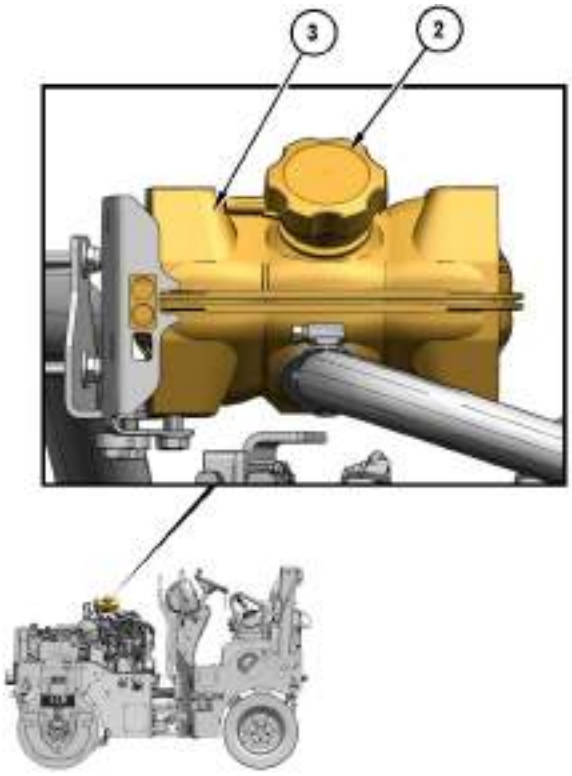


Illustration 97 g06675368

(2) Pressure cap
(3) Shunt tank

2. Loosen the cooling system pressure cap (2) slowly to relieve pressure. Refer to “System Pressure Release“ for more information. Remove the cooling system pressure cap (2).

3. Inspect the cooling system pressure cap (2) for foreign material, for deposits, and for damage. Clean the cooling system pressure cap (2) with a clean cloth. If the cooling system pressure cap (2) is damaged, replace the cooling system pressure cap (2).

4. Install the cooling system pressure cap (2).

5. Close the engine compartment hood (1). Refer to “Access Doors and Covers“ for more information.

i08468409

Cooling System Water Temperature Regulator - Replace

SMCS Code: 1355-510; 1393-010

WARNING

At operating temperature, the engine coolant is hot and under pressure.

Steam can cause personal injury.

Check the coolant level only after the engine has been stopped and the fill cap is cool enough to touch with your bare hand.

Remove the fill cap slowly to relieve pressure.

Cooling system conditioner contains alkali. Avoid contact with the skin and eyes to prevent personal injury.

WARNING

Pressurized System: Hot coolant can cause serious burns. To open the cooling system filler cap, stop the engine and wait until the cooling system components are cool. Loosen the cooling system pressure cap slowly in order to relieve the pressure.

NOTICE
Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting, and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, PERJ1017, “Dealer Service Tool Catalog” for tools and supplies suitable to collect and contain fluids on Cat ® products.

Dispose of all fluids according to local regulations and mandates.

NOTICE
Failure to replace the engine's water temperature regulator on a regularly scheduled basis could cause severe engine damage.

NOTICE
The water temperature regulators may be reused if the water temperature regulators are within test specifications, are not damaged, and do not have excessive buildup of deposits.

NOTICE
Since Caterpillar engines incorporate a shunt design cooling system, it is mandatory to always operate the engine with a water temperature regulator.

Depending on load, failure to operate with a water temperature regulator could result in either an overheating or an overcooling condition.

NOTICE
If the water temperature regulator is installed incorrectly, it will cause the engine to overheat.

NOTICE
Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

NOTICE
Not following the recommendations found in this manual can lead to reduced performance and compartment failure.

Prepare the machine for maintenance. Refer to “Prepare the Machine for Maintenance“ for more information.

Replace the water temperature regulator regularly to reduce the chance of unscheduled downtime and of problems with the cooling system.

The water temperature regulator should be replaced after the cooling system has been cleaned. Replace the water temperature regulator while the cooling system is completely drained. Replace the water temperature regulator while the cooling system coolant is drained to a level below the water temperature regulator housing.

Note: If you are only replacing the water temperature regulator, drain the cooling system coolant to a level that is below the water temperature regulator housing.

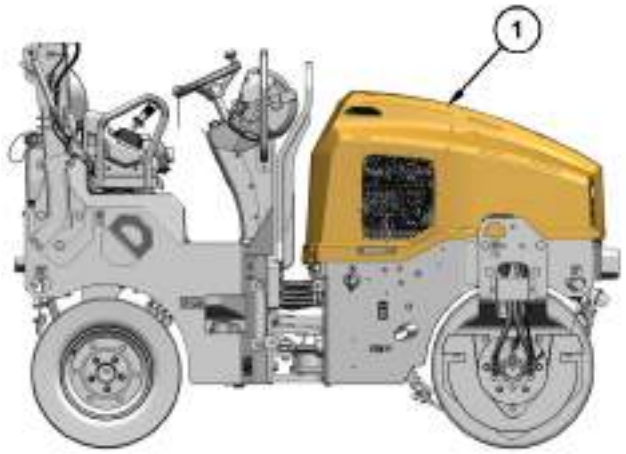


Illustration 98 g06674584

(1) Hood

1. Open the engine compartment hood (1). Refer to “Access Doors and Covers“ for more information.

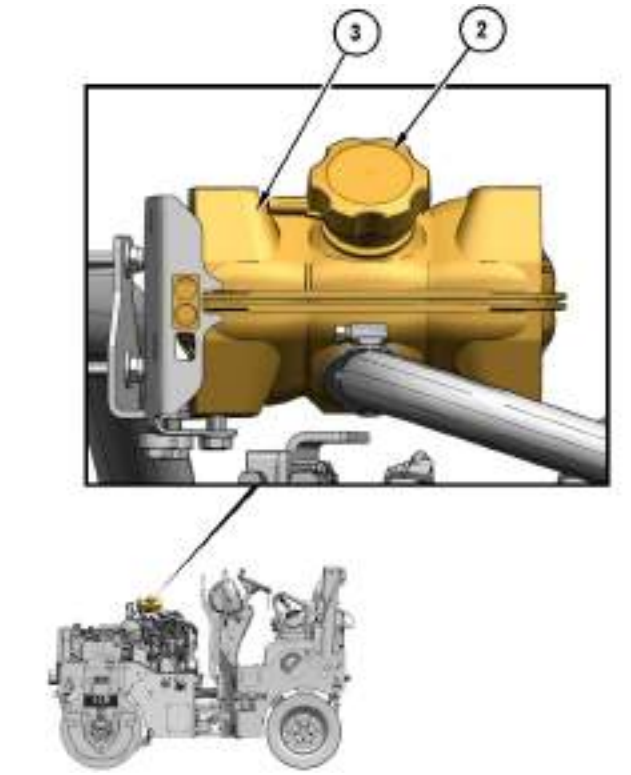


Illustration 99 g06675368

Some parts removed for clarity

(2) Pressure cap
(3) Shunt tank

2. Loosen the cooling system pressure cap (2) to relieve the pressure in the cooling system. Refer to “System Pressure Release“ for more information. Remove the cooling system pressure cap (2).

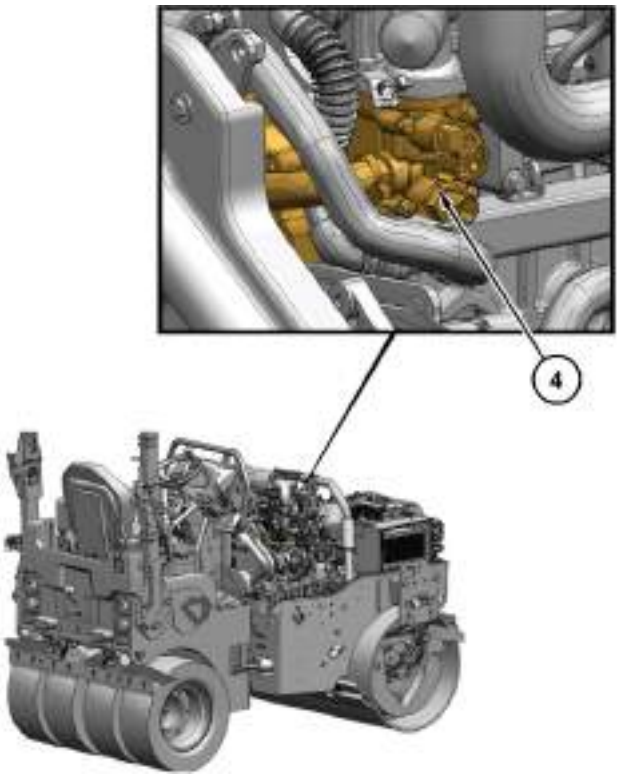


Illustration 100 g06675582

Some parts removed for clarity

(4) Water temperature regulator

3. Remove the housing for water temperature regulator (4).

4. Remove the gasket and remove water temperature regulator (4).

5. Install new water temperature regulator (4) and install a new gasket.

6. Install the housing for water temperature regulator (4).

Note: Refer to “General Hazard Information“ for more information that pertains to containing fluid spillage.

7. Add the cooling system coolant. Maintain the level of the coolant to 10 mm (0.4 inch) from the bottom of the fill pipe.

8. Inspect cooling system pressure cap (2) and the gasket for damage. Replace pressure cap (2) if pressure cap (2) or the gasket is damaged.

9. Install the cooling system pressure cap (2).

10. Close the engine compartment hood (1). Refer to “Access Doors and Covers“ for more information.

i08470251

Drum Scrapers - Inspect/ Adjust/Replace

SMCS Code: 6607-510; 6607-025; 6607-040

WARNING

Block the machine in order to prevent unexpected movement of the machine. Failure to do so may result in personal injury or death.

Prepare the machine for maintenance. Refer to “Prepare the Machine for Maintenance“.

Inspect Scrapers

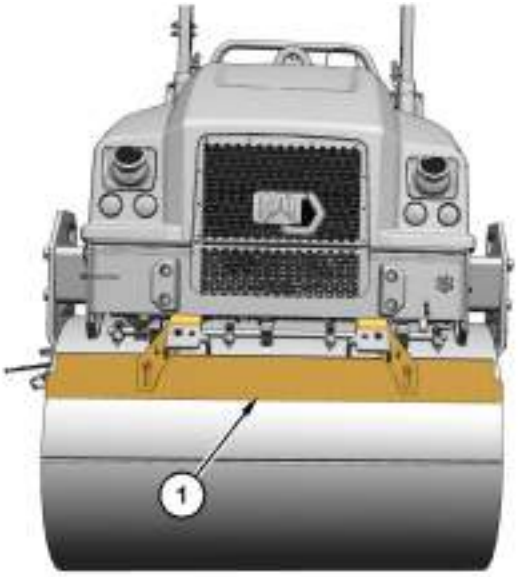


Illustration 101 g06676657

Front view of the machine

(1) Scraper

1. Remove dirt and debris from scraper (1).

2. Inspect scraper (1) for wear or damage.

3. Move scraper (1) down using the adjustment slots that are provided on the scraper support, to maintain the original angle of the scraper. If there is no space remaining in the slot for further adjustment, replace scraper (1).

Adjust Scrapers

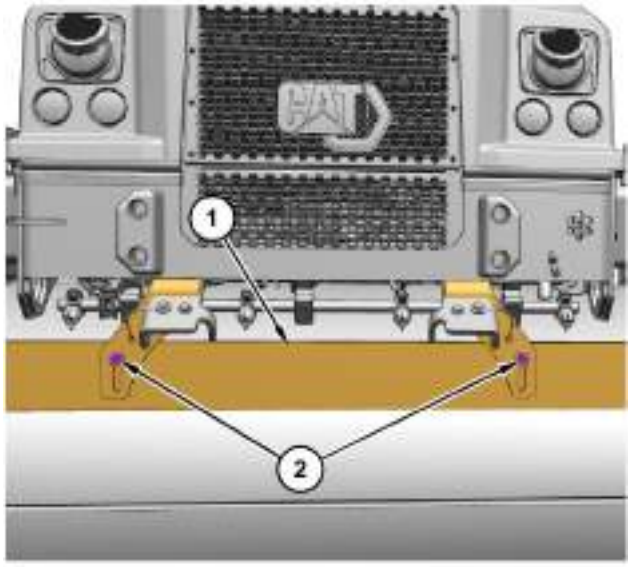


Illustration 102 g06676729

Front view of the machine

(1) Scraper
(2) Bolt

1. Adjust scraper (1) so that scraper (1) touches the width of the drum.

2. Loosen bolts (2) to adjust the scraper. When scraper (1) is in the desired position, tighten bolts (2). Refer to Specifications, SENR3130, “Torque Specifications”.

Replace Scrapers

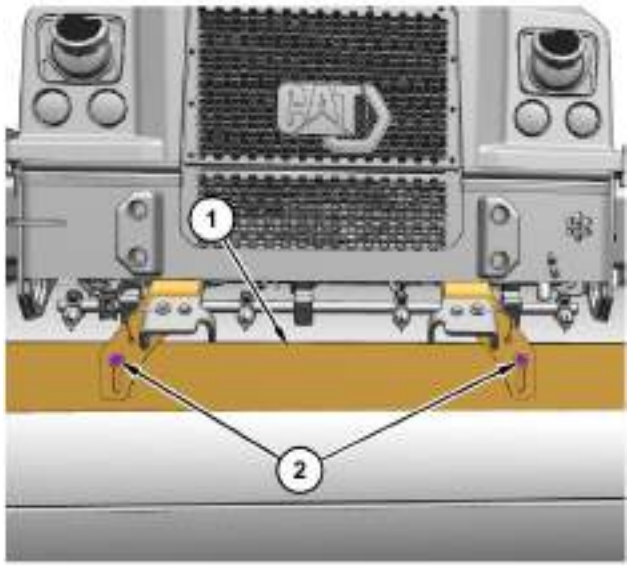


Illustration 103 g06676729

Front view of the machine

(1) Scraper
(2) Bolt

1. Loosen bolts (2) and remove damaged scraper (1).

2. Install new scraper (1) and tighten bolts (2). Refer to Specifications, SENR3130, “Torque Specifications”.

3. Adjust scraper (1) to touch the drum.

4. Install scraper (1) so that bolts (2) are at the top of the adjustment slots in the scraper support.

Replace Cocoa Mat

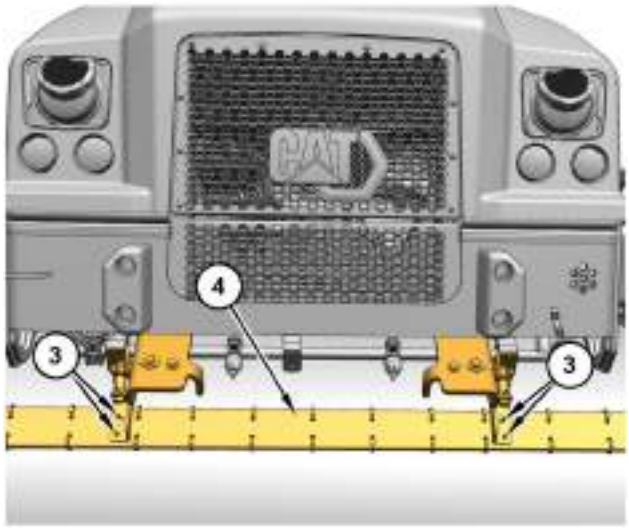


Illustration 104 g06676924

Front view of the machine

(3) Bolt
(4) Cocoa mat

1. Loosen bolts (3) and remove damaged cocoa mat (4).

2. Install new cocoa mat (4) and tighten bolts (3). Refer to Specifications, SENR3130, “Torque Specifications”.

3. Adjust cocoa mat (4) using bolts (3), so that the cocoa mat (4) touches the drum.

i06250385

Eccentric Weight Bearings - Inspect

SMCS Code: 6606-040-BD

The eccentric weight bearings should be inspected periodically as part of a preventive maintenance schedule. Consult your Cat dealer for information concerning the inspection of the eccentric weight bearings.

For additional information, refer to Disassembly and Assembly, “Eccentric Weight - Remove and Install”.

i07819515

Engine - Clean

SMCS Code: 1000-070; 1000

⚠ WARNING

Personal injury or death can result from high voltage.

Moisture can create paths of electrical conductivity.

Make sure that the electrical system is OFF. Lock out the starting controls and tag the controls “DO NOT OPERATE” .

NOTICE
Accumulated grease and oil on an engine is a fire hazard. Keep the engine clean. Remove debris and fluid spills whenever a significant quantity accumulates on the engine.

Periodic cleaning of the engine is recommended. Steam cleaning the engine will remove accumulated oil and grease. A clean engine provides the following benefits:

- Easy detection of fluid leaks
- Maximum heat transfer characteristics
- Ease of maintenance

Note: Caution must be used to prevent electrical components from being damaged by excessive water when the engine is cleaned. Pressure washers and steam cleaners should not be directed at any electrical connectors or the junction of cables into the rear of the connectors. Avoid electrical components such as the alternator, the starter, and the ECM. Protect the fuel injection pump from fluids to wash the engine.

Ensure that care is taken that the safety labels, emission label, and all other information labels are not removed during engine cleaning.